

X_1	X_2	X_3	Y
1	1	0	1
1	0	0	1
0	1	1	0
0	0	1	0
1	1	1	0

Using Entropy and Information Gain: (1) Compute the entropy of the root node. (2) Compute the information gain for splits on X_1 and X_3 . (3) Select the best feature for the root node and construct the resulting decision tree.

Part B: CLO 2

(Attempt ALL THREE questions from this section)

Q6 (CLO 2) [10 Marks]

You are given the one-dimensional dataset $X = \{1, 2, 8, 9\}$. Assuming $k = 2$, initialize the first centroid as $c_1 = 1$, and select the second centroid using the K-means++ strategy. Perform exactly one complete iteration of the K-means algorithm to find the updated centroids.

Q7 (CLO 2) [10 Marks]

Given the 1D data points 18, 22, 25, 27, 42, 43, perform agglomerative clustering using single linkage (nearest neighbor). Show the sequence of cluster merges alongside their minimum distances, and illustrate your final result as a dendrogram.

Q8 (CLO 2) [10 Marks]

We are monitoring the response time (in milliseconds) of a web server. The observed normal data is $X = \{20, 22, 21, 23\}$. Assuming the data follows a Gaussian (normal) distribution, evaluate whether a new observation $x = 35$ is an anomaly using an anomaly threshold of $\epsilon = 0.005$. Justify your answer mathematically by computing the necessary distribution parameters and probability density. ~~Briefly comment on whether odd or even values of k are preferable.~~

Hint: The probability density function for a Gaussian distribution is given by:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Part A: CLO 1
(Attempt any THREE questions from this section)

Q1 (CLO 1) [10 Marks]

You are given the one-dimensional dataset:

$$\{(1, 0), (2, 0), (3, 0), (5, 1), (8, 1)\}$$

For a query point located at $x = 4$, determine its predicted class using the k -Nearest Neighbors algorithm for both $k = 2$ and $k = 4$. Compare the results and briefly discuss the role of k in classification stability. (Note: In case of a tie, break the tie by choosing the class of the closest neighbor. If distances are also equal, choose the class with the smaller label.)

Q2 (CLO 1) [10 Marks]

Consider the loss function with a regularization parameter λ :

$$L(w) = (w - 2)^2 + \lambda w^2$$

Derive the gradient descent update rule for a learning rate η , and briefly explain how the regularization term λw^2 helps prevent overfitting.

Q3 (CLO 1) [10 Marks]

The hard-margin Support Vector Machine (SVM) optimization problem is defined as:

$$\min_{w, b} \frac{1}{2} \|w\|^2 \quad \text{subject to} \quad y_i(w^T x_i + b) \geq 1$$

Explain the geometric meaning of this formulation. Specifically, discuss how the objective function and the constraint jointly determine the margin between classes.

Q4 (CLO 1) [10 Marks]

The dual form of SVM maximizes $\sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j (x_i^T x_j)$. Explain why the SVM formulation depends only on dot products $x_i^T x_j$. Then explain how replacing these dot products with a kernel function allows SVM to learn non-linear decision boundaries. Illustrate your answer using any kernel of your choice.

Q5 (CLO 1) [10 Marks]

You are given the following dataset with binary features: